

THE BLACK-SCHOLES MARKET

JOSH MACDONALD

Purpose of the Project

The aim of this project is to implement Black–Scholes option pricing theory in Python. This work is being done for my own personal development and enjoyment, but also to strengthen my understanding of how mathematical models can be translated into working financial code.

These notes act as a complementary guide to the project. They outline the main theory behind the Black–Scholes model, derive the pricing formula for European call and put options, and explain how the mathematical formula is implemented computationally. In addition, there may be further sections introducing other type contingent claims.

It should be noted that these notes do not cover all of the probability and measure theory required for a fully rigorous treatment of the Black–Scholes model. In particular, topics such as Brownian motion, stochastic integration, martingales, filtrations and equivalent martingale measures are only discussed at the level needed for this project. A selection of textbooks are included at the end which act as references but the main content is adapted from the course in probability, measure and finance from the university of Edinburgh taught by Leonardo Tolomeo.

The Black–Scholes Model

To begin, we will work on a filtered probability space, which is given as:

$$(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P}),$$

supporting a Brownian motion $W = (W_t)_{t \geq 0}$ and the filtration $(\mathcal{F}_t)_{t \geq 0}$ represents the information available to the market up to time t .

In the Black–Scholes model, the market consists of two assets:

- a risky asset, usually interpreted as a stock;
- a riskless asset, usually interpreted as a bank account or bond.

The stock price S_t follows the stochastic differential equation (SDE):

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad S_0 > 0, \quad (1)$$

where μ is the drift (expected rate of return) and $\sigma > 0$ is the volatility.

The riskless asset B_t satisfies

$$dB_t = r B_t dt, \quad B_0 = 1, \quad (2)$$

where r is the continuously compounded risk-free interest rate.

In the classical Black–Scholes model, μ , r and σ are assumed to be constant. Under these assumptions, the solutions are:

$$S_t = S_0 \exp \left(\left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma W_t \right), \quad (1A)$$

and

$$B_t = e^{rt}. \quad (2A)$$

The stock price is therefore lognormally distributed. This is one of the key reasons why the Black–Scholes formula can be written in closed form.

Definition 0.1 (numéraire). A numéraire is a strictly positive price process used as a unit of account. If X_t is an asset price and $N_t > 0$ is a numéraire, then the discounted or relative price of X_t in units of N_t is:

$$\frac{X_t}{N_t}.$$

Using the bank account $B_t = e^{rt}$ as numéraire, the discounted stock price, $\tilde{S}_t$, is:

$$\tilde{S}_t = \frac{S_t}{B_t} = e^{-rt} S_t.$$

Discounted Prices and No-Arbitrage Intuition

It is convenient to work with discounted prices because it expresses all values in units of the riskless asset. That is, instead of measuring wealth in pounds, we measure wealth relative to the bank account B_t . This removes the deterministic growth due to the risk-free interest rate.

We now introduce one of the most important formulas in stochastic calculus, Itô's formula, for which we first define an Itô process:

Definition 0.2 (Itô process). An Itô process is a stochastic process $X = (X_t)_{t \geq 0}$ that can be written in the following form,

$$X_t = X_0 + \int_0^t a_s ds + \int_0^t b_s dW_s,$$

where W_t is a Brownian motion and a_t, b_t are adapted processes satisfying suitable integrability conditions, i.e for every $T > 0$, we have that $\int_0^T |a_s|^2 ds < \infty$ and $\int_0^T |b_s|^2 ds < \infty$ a.s.

As well as this, we equivalently have in differential notation the following:

$$dX_t = a_t dt + b_t dW_t.$$

The term a_t is called the drift coefficient, and b_t is called the diffusion coefficient.

Theorem 0.3 (Itô's formula). Let X_t be an Itô process satisfying

$$dX_t = a_t dt + b_t dW_t,$$

where a_t and b_t are adapted processes. If $f \in C^2(\mathbb{R})$, then $f(X_t)$ is also an Itô process and satisfies

$$df(X_t) = f'(X_t) dX_t + \frac{1}{2} f''(X_t) (dX_t)^2.$$

Equivalently,

$$df(X_t) = \left(a_t f'(X_t) + \frac{1}{2} b_t^2 f''(X_t) \right) dt + b_t f'(X_t) dW_t.$$

Remark 0.4. The additional second-order term

$$\frac{1}{2} b_t^2 f''(X_t) dt$$

is the key difference between Itô calculus and ordinary calculus. It appears because Brownian motion has quadratic variation $(dW_t)^2 = dt$.

Now we have Itô's formula, we can apply it to the discounted stock price, which gives us:

$$d\tilde{S}_t = d(e^{-rt} S_t).$$

Since

$$d(e^{-rt}) = -re^{-rt} dt,$$

we obtain

$$d\tilde{S}_t = e^{-rt} dS_t - re^{-rt} S_t dt.$$

Substituting in the Black–Scholes stock dynamics

$$dS_t = \mu S_t dt + \sigma S_t dW_t,$$

gives

$$d\tilde{S}_t = e^{-rt}(\mu S_t dt + \sigma S_t dW_t) - r e^{-rt} S_t dt.$$

Therefore,

$$d\tilde{S}_t = (\mu - r)\tilde{S}_t dt + \sigma \tilde{S}_t dW_t.$$

This shows that the discounted stock price still has a drift term under the original probability measure $\mathbb{P}$, unless $\mu = r$. In general, the real-world expected return μ is not equal to the risk-free rate r .

The key idea behind risk-neutral pricing is that we can move to a new probability measure $\mathbb{Q}$, called the risk-neutral measure, under which the discounted stock price has no drift. Under this measure,

$$d\tilde{S}_t = \sigma \tilde{S}_t dW_t^{\mathbb{Q}}.$$

Thus, under $\mathbb{Q}$, the discounted stock price is a martingale. This means that its expected future discounted value is equal to its current value:

$$\tilde{S}_t = \mathbb{E}^{\mathbb{Q}}[\tilde{S}_T | \mathcal{F}_t].$$

This martingale property is central to modern derivative pricing. It means that once prices are discounted by the riskless asset, there should be no predictable way to make a risk-free profit. This is closely related to the no-arbitrage principle.

Self-Financing Portfolios

A trading strategy is a choice of how much wealth to hold in the stock and how much to hold in the riskless asset. Suppose a portfolio holds Δ_t units of the stock and β_t units of the bank account. The value of the portfolio is

$$V_t = \Delta_t S_t + \beta_t B_t.$$

The portfolio is called self-financing if changes in its value come only from changes in the prices of the assets already held. That is, no money is added to or withdrawn from the portfolio after it is set up.

Mathematically, a self-financing portfolio satisfies

$$dV_t = \Delta_t dS_t + \beta_t dB_t.$$

This idea is important because the Black–Scholes argument is based on replicating a derivative payoff using a self-financing trading strategy. If a portfolio can be constructed so that

$$V_T = h(S_T),$$

then, by no-arbitrage, the price of the derivative at time 0 must be equal to the initial cost V_0 of the replicating portfolio.

If this were not true, then an arbitrage opportunity would exist. For example, if the derivative were cheaper than the replicating portfolio, one could buy the derivative and sell the portfolio. If it were more expensive, one could sell the derivative and buy the portfolio. In either case, this would create a risk-free profit, which is ruled out by the no-arbitrage assumption.

Risk-Neutral Pricing

The Black–Scholes model prices derivatives by using a risk-neutral probability measure, usually denoted by $\mathbb{Q}$. Under this measure, the discounted stock price is a martingale. Informally, this means that after discounting by the risk-free rate, the stock earns no additional expected return under $\mathbb{Q}$.

Under the risk-neutral measure, the stock price dynamics become

$$dS_t = rS_t dt + \sigma S_t dW_t^{\mathbb{Q}},$$

where $W_t^{\mathbb{Q}}$ is a Brownian motion under $\mathbb{Q}$. The solution is

$$S_T = S_0 \exp\left(\left(r - \frac{1}{2}\sigma^2\right)T + \sigma W_T^{\mathbb{Q}}\right).$$

Therefore, for a European option with payoff $h(S_T)$, the no-arbitrage price at time 0 is

$$V_0 = e^{-rT} \mathbb{E}^{\mathbb{Q}}[h(S_T)].$$

This formula is the theoretical foundation of the Black–Scholes pricing formula.

European Type Contingent Claims

A contingent claim is a financial contract whose value depends on the value of some underlying asset. In this project, the underlying asset is the stock price S_t .

A European option is a contract that can only be exercised at maturity T . This is different from an American option, which can be exercised at any time $t \in [0, T]$.

A European-type contingent claim has payoff

$$h = g(S_T),$$

where g is a non-negative Borel measurable function.

For a European call option with strike price K , the payoff is

$$C_T = \max(S_T - K, 0) = (S_T - K)^+.$$

For a European put option with strike price K , the payoff is

$$P_T = \max(K - S_T, 0) = (K - S_T)^+.$$

The call option gives the holder the right, but not the obligation, to buy the stock at price K at maturity. The put option gives the holder the right, but not the obligation, to sell the stock at price K at maturity.

American Type Contingent Claims

Placeholder

Asian and Exotic Type Contingent Claims

placeholder

The Black–Scholes Pricing Formula

The Black–Scholes formula gives closed-form prices for European call and put options. It is obtained by applying the risk-neutral valuation formula to the payoff of a European option.

Under the risk-neutral measure $\mathbb{Q}$, the stock price satisfies

$$dS_t = rS_t dt + \sigma S_t dW_t^{\mathbb{Q}}.$$

Therefore,

$$S_T = S_0 \exp \left(\left(r - \frac{1}{2} \sigma^2 \right) T + \sigma W_T^{\mathbb{Q}} \right).$$

Since $W_T^{\mathbb{Q}} \sim N(0, T)$, we can write

$$W_T^{\mathbb{Q}} = \sqrt{T}Z, \quad Z \sim N(0, 1).$$

Hence

$$S_T = S_0 \exp \left(\left(r - \frac{1}{2} \sigma^2 \right) T + \sigma \sqrt{T}Z \right).$$

This shows that S_T is lognormally distributed under the risk-neutral measure.

Theorem 1.1 (Black–Scholes formula). *In the Black–Scholes model, the time-zero price of a European call option with strike K and maturity T is*

$$C = S_0 N(d_1) - K e^{-rT} N(d_2),$$

and the time-zero price of a European put option with the same strike and maturity is

$$P = K e^{-rT} N(-d_2) - S_0 N(-d_1),$$

where

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

Here $N(x)$ denotes the cumulative distribution function of the standard normal distribution:

$$N(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-z^2/2} dz.$$

Sketch of the call option derivation. The payoff of a European call option is

$$(S_T - K)^+.$$

By risk-neutral valuation, its time-zero price is

$$C = e^{-rT} \mathbb{E}^{\mathbb{Q}} [(S_T - K)^+].$$

This can be written as

$$C = e^{-rT} \mathbb{E}^{\mathbb{Q}} [(S_T - K) \mathbf{1}_{\{S_T > K\}}].$$

Expanding the expectation gives

$$C = e^{-rT} \left(\mathbb{E}^{\mathbb{Q}} [S_T \mathbf{1}_{\{S_T > K\}}] - K \mathbb{Q}(S_T > K) \right).$$

Using the lognormal distribution of S_T , one obtains

$$\mathbb{Q}(S_T > K) = N(d_2),$$

and

$$e^{-rT} \mathbb{E}^{\mathbb{Q}} [S_T \mathbf{1}_{\{S_T > K\}}] = S_0 N(d_1).$$

Therefore,

$$C = S_0 N(d_1) - K e^{-rT} N(d_2).$$

The put option formula follows either by a similar calculation or from put–call parity. □

Remark 1.2. The Black–Scholes formula does not depend on the real-world expected return μ . This is because the option is priced under the risk-neutral measure $\mathbb{Q}$, where the stock has drift r , not μ . The inputs to the formula are therefore

$$S_0, \quad K, \quad T, \quad r, \quad \sigma.$$

Remark 1.3 (Interpretation of d_1 and d_2). The quantities d_1 and d_2 measure the moneyness of the option after adjusting for interest rates, time to maturity and volatility.

The term $N(d_2)$ can be interpreted as the risk-neutral probability that the call option finishes in the money:

$$N(d_2) = \mathbb{Q}(S_T > K).$$

The term $N(d_1)$ is related to the sensitivity of the call price with respect to the current stock price. In fact, for a European call option, $N(d_1)$ is the option's Delta:

$$\Delta_{\text{call}} = \frac{\partial C}{\partial S_0} = N(d_1).$$

Put–Call Parity

Put–call parity is an important relationship between European call and put prices with the same strike price K and maturity T . It is given by

$$C - P = S_0 - Ke^{-rT}.$$

Equivalently,

$$C + Ke^{-rT} = P + S_0.$$

This identity is useful for checking the correctness of the implementation. If the Python program returns call and put prices that do not approximately satisfy put–call parity, then there is likely an error in the calculation.

Implementation Details

The Python program takes the following inputs:

$$S_0, \quad K, \quad T, \quad r, \quad \sigma.$$

These represent:

Symbol	Meaning
S_0	Initial stock price
K	Strike price
T	Time to maturity
r	Risk-free interest rate
σ	Volatility of the stock

The program then follows these steps:

- (1) Compute d_1 :

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}.$$

- (2) Compute d_2 :

$$d_2 = d_1 - \sigma\sqrt{T}.$$

- (3) Evaluate the standard normal cumulative distribution function at d_1 , d_2 , $-d_1$ and $-d_2$.

- (4) Return the call price:

$$C = S_0N(d_1) - Ke^{-rT}N(d_2).$$

(5) Return the put price:

$$P = Ke^{-rT}N(-d_2) - S_0N(-d_1).$$

In Python, the normal cumulative distribution function can be computed using `scipy.stats.norm.cdf`. The exponential and logarithmic terms can be computed using either the standard `math` library or `numpy`.

Numerical Example

We consider the following usual example, which is taken from :

$$S_0 = 100, \quad K = 100, \quad T = 1, \quad r = 0.05, \quad \sigma = 0.2.$$

Then

$$d_1 = \frac{\ln(100/100) + (0.05 + \frac{1}{2}(0.2)^2)(1)}{0.2\sqrt{1}} = 0.35,$$

and

$$d_2 = 0.35 - 0.2 = 0.15.$$

The Black-Scholes call price is approximately

$$C \approx 10.4506,$$

and the put price is approximately

$$P \approx 5.5735.$$

We can also verify put-call parity:

$$C - P \approx 10.4506 - 5.5735 = 4.8771,$$

while

$$S_0 - Ke^{-rT} = 100 - 100e^{-0.05} \approx 4.8771.$$

Therefore, the numerical output is consistent with put-call parity.

Ideas for Further Work

This project implements the Black-Scholes pricing formula for European call and put options. The mathematical theory begins with modelling the stock price as a geometric Brownian motion and using risk-neutral pricing to compute the discounted expected payoff of an option.

Possible extensions to this project include adding dividend yields, pricing American Options, Exotic Options, calculating Greeks, plotting option prices as functions of input parameters, and comparing Black-Scholes prices with Monte Carlo estimates.

Another idea was to implement a nice-looking interface for the option calculations.

References

- Björk, T. *Arbitrage Theory in Continuous Time*. Oxford University Press.
- Shreve, S. *Stochastic Calculus for Finance II: Continuous-Time Models*. Springer.
- Hull, J. *Options, Futures, and Other Derivatives*. Pearson.